

p and q are different prime numbers. r is the least prime number greater than p , and s is the least prime number greater than q .

6. $r - p$ $s - q$

4

Let $p=2$ and $q=3$
and $r=3$ and $s=5$
 $3-2 < 5-3$
 $1 < 2$
similarly, if $p=3$ and $q=2$
and $r=5$ and $s=3$
 $5-3 > 3-2$
 $2 > 1$
so D is the right choice

26. If $0 < st < 1$, then which of the following can be true?

- (A) $s < -1$ and $t > 0$
- (B) $s < -1$ and $t < -1$
- (C) $s > -1$ and $t < -1$
- (D) $s > 1$ and $t < -1$
- (E) $s > 1$ and $t > 1$

st, both has to be negative or positive fraction.

Either s or t has to negative integer and another negative fraction

a) $t > 0$ and $s < -1$ since st is negative. it is wrong answer.

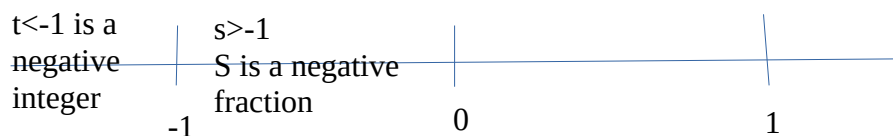
D) $s > 1$ and $t < -1$ since st is negative. it is wrong ansewr.

E) $s > 1$ and $t > 1$ since st is not a fraction, it is integer .it is wrong answer.

Now,

b) $s < -1$ and $t < -1$. then st is not fraction rather integer.

*c) $s > -1$ and $t < -1$.then st is a positive fraction. **It is correct.***



st is positive fraction

GIVEN, $[d]$ is the rounded expression of d .

if $x=5038.94$

$\text{rounded}(5038.94)=5039 \dots (i)$

$d = 5.03894$ rounded to nearest thousand

$5.03894 = (5038.94/1000) \dots (ii)$

from (I) $5038.94 = 5039$ (after rounding) from (i)

now from (i)

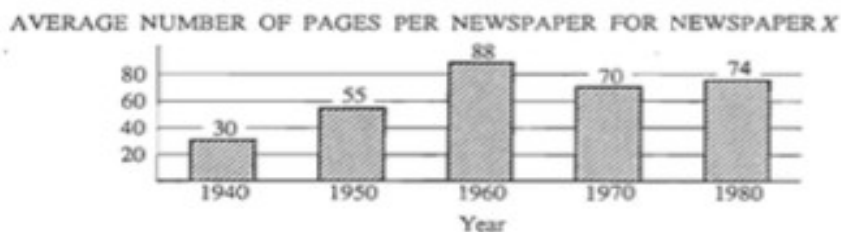
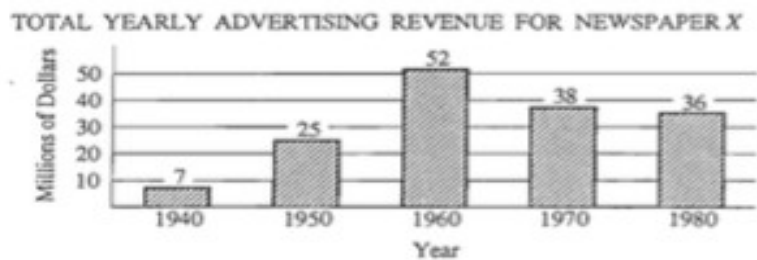
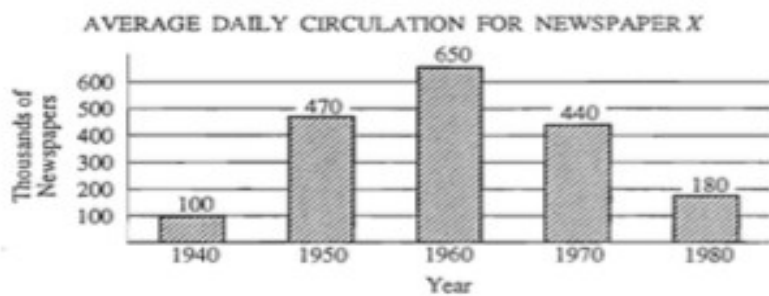
$(5039/1000) = 5.039$

$[d] = 5.039$

decimal point $d(5.03894)$ and $[d](5.039)$ differ $= 3$ (894)

so b is answer.

Questions 21-25 refer to the following graphs.



21) In how many of the years shown was the average number of pages per newspaper at least twice as much as the average in 1940?

figure 3

number of pages per newspaper in 1940 $\Rightarrow 30$
 $30 \times 2 = 60$

In how many of the years shown was the average number of pages per newspaper at least 70?

1960, 1970, 1980 $\Rightarrow 3$ years

21.B

22.

printing cost = .05

$(470 \times 1000 \times 5) / 100$

23500

Answer c

23. 1980 total revenue = 36

$$36 = x \cdot 180$$

$$x = 36 \cdot 10^6 / 180 \cdot 10^3$$

$$= 200$$

23)c is the correct answer.

24)

$$1960 = 650$$

$$1970 = 440$$

$$((650 - 440) \cdot 100) / 650$$

$$= 32\%$$

24)E.

$$25) 1950 \Rightarrow 25 \text{ and } 1950 \Rightarrow 52$$

$$52 - 25 = 27$$

i) correct

ii) according to figure 1 and figure 2.

since two graphs are similar

(ii) is correct.

$$\text{Iii) } 1970 \Rightarrow 70$$

$$1980 \Rightarrow 74$$

$$74 - 70 = 4$$

iii) wrong answer.

25.D is correct answer